

Reflective Journal

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For this midterm, our team worked with the Stroke Prediction Dataset to clean and preprocess it for machine learning. The biggest challenge going in was learning how to balance everything out the right way, using code instead of editing the data manually, and making sure we only balanced the training data without ever touching the test set.

One thing our team did not expect was how much small coding issues could slow us down. Something as minor as a typo, like a misspelled column name or a wrong variable, could stop a whole cell from running, and it took some patience to track those down and fix them. It taught us that in machine learning the details really matter, and that being careful and precise with the code is just as important as understanding the concepts behind it.

A big part of this project was coordinating as a team through each step. We had to make sure everyone understood the order of the pipeline and stayed on the same page about what needed to happen and when. Keeping our communication clear in the group chat helped us avoid mistakes and made sure the work actually came together instead of everyone pulling in different directions.

The most important rule we learned was what our instructor called the golden rule. We had to split the data first, then only balance the training set, and never touch the test set. Using SMOTE, we added synthetic stroke cases to the training data so the model would have enough examples to learn from, while leaving the test set at its real proportions. We learned that if you balance the data before splitting, or apply SMOTE to the test set, you corrupt the whole experiment and the results can no longer be trusted.

Our team also gained a real appreciation for how much work goes into preparing raw data before any model can use it. The dataset had missing values, categories that needed to be converted into numbers, and features on very different scales, and all of that had to be handled through code before the data was usable. It showed us that data preparation is a huge part of machine learning and not just an afterthought.

Our team is interested in a lot of different subjects, but healthcare stood out to us as an area where machine learning can make a real difference. This feels especially relevant living in Houston, home to one of the largest medical centers in the world. A project like predicting strokes shows how these tools could actually support doctors and patients, and that made the work feel meaningful beyond just completing an assignment.